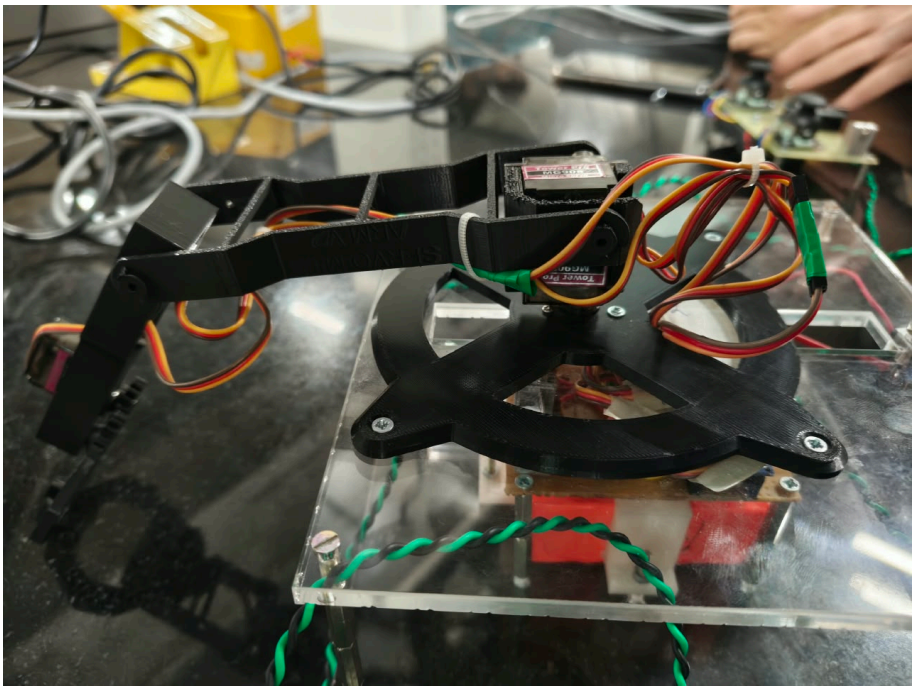
pinchy assembly guide

by mwezi2000

hey! this is how i put my robot arm together, in the same order i actually did it. its a 4 dof arm (base, shoulder, elbow and a claw) that you control with 2 joysticks. the remote has an arduino nano that reads the joysticks and sends the values over a 2 wire cable to an arduino uno in the base, and the uno moves the 4 servos.

dont skip the testing part, its way easier to find a bad solder joint when the board is still on your desk and not stuck under the base lol



the finished arm

what you need

part	how many
arduino nano	1
arduino uno	1
hw504 joystick	2
mg90s servo (metal gear)	4
lm7805 regulator	4
0.01uF and 0.1uF ceramic caps	4 each
1n4007 or 1n5819 diode	2
spst toggle switch	2
red led and 330 ohm resistor	2 each
7.4v 2s li-ion battery (remote)	1
11.1v 3s battery pack (arm)	1

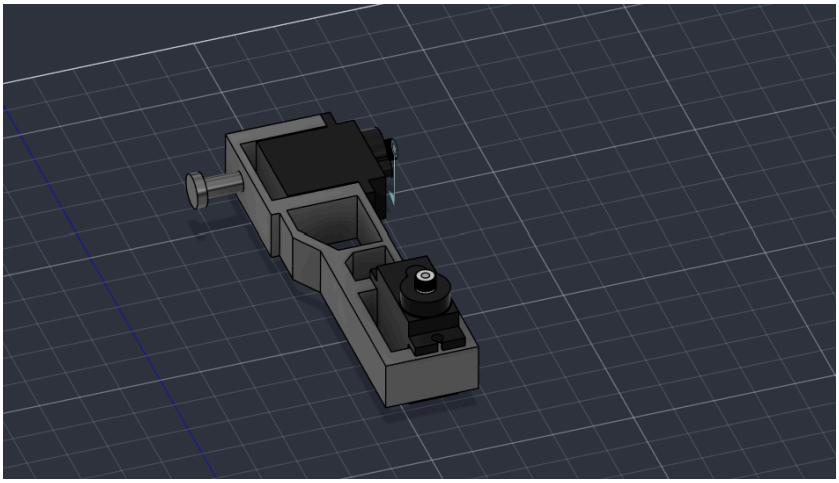
zero pcb, 8x18 cm and 10x10 cm	1 each
female berg strips, 2 pin connectors	a few
m3 bolts, nuts and spacers	lots
twisted 2 core wire	about 1 m
pla+ filament	about 70 g
3-5 mm acrylic sheet	1
zip ties	a bunch

tools: soldering iron, multimeter (you really need this one), screwdrivers, cutters, wire stripper, 3d printer, laser cutter (or get the acrylic cut somewhere) and a laptop with the arduino ide.

step 1: print and cut the parts

print the base plate, servo bracket, arm 1, arm 2 and the gripper parts (both gears, the fingers and the top cover) from the step files in the CAD folder. i used black pla+ with normal settings, just make sure arm 2 is printed solid since it carries the most weight. then laser cut the bottom base from the acrylic.

i left 0.2-0.3 mm gap on all the servo pockets and screw holes. if a servo still doesnt slide in, file the pocket a little bit. dont force it, the plastic cracks.



arm 2 in fusion, you can see the pockets for the servos

step 2: solder the remote

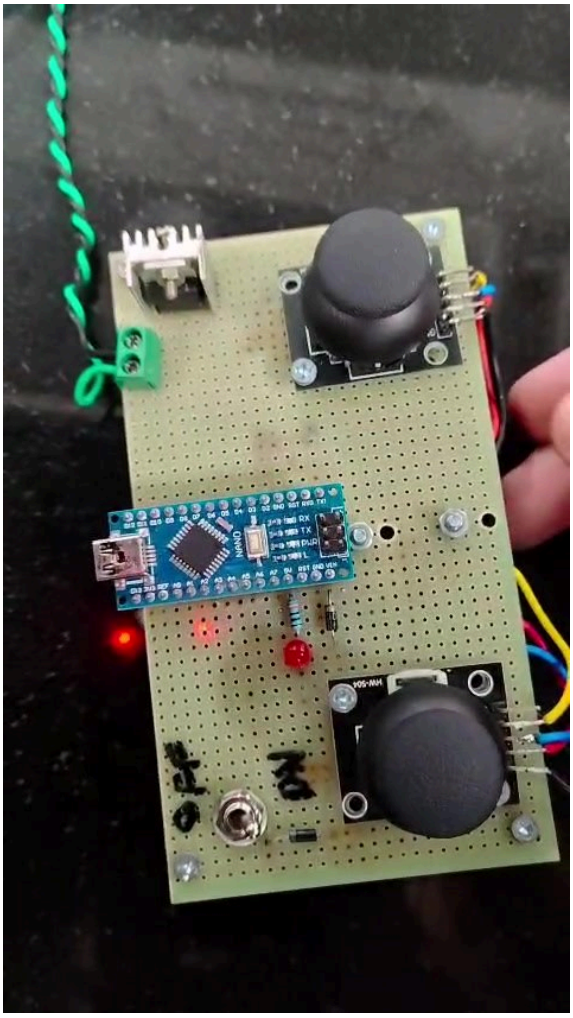
start with this one, its the easier board. parts go on top and the jumper wires go on the bottom. plan where everything goes on paper first so youre not guessing halfway. i put the joysticks on 5 pin female headers instead of soldering them straight on, so i can swap them if one dies.

connections:

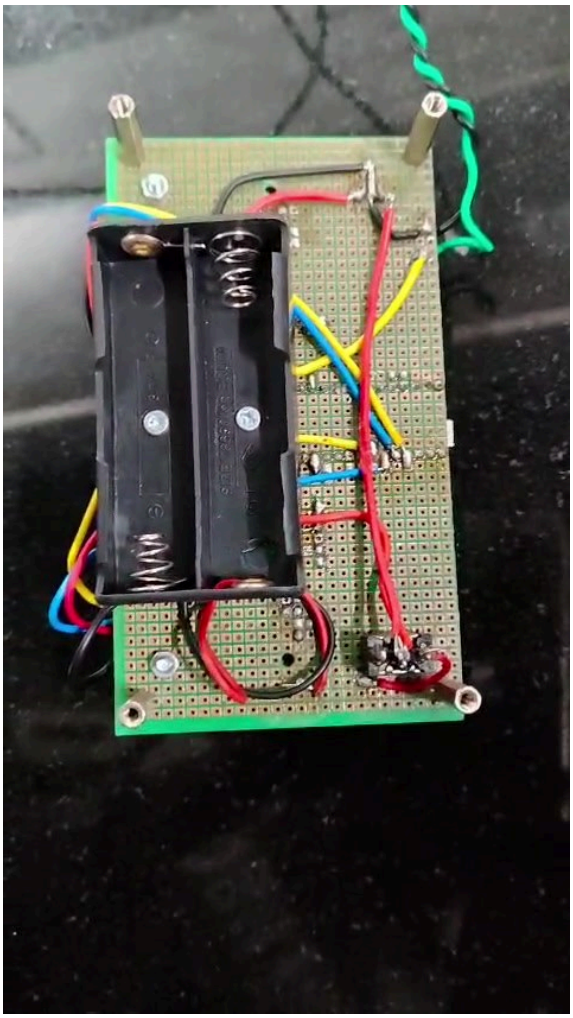
- battery + to diode, then the switch, then nano VIN
- battery - to nano GND
- nano 5V to both joysticks +5V, and to the led through the 330 ohm resistor
- joystick 1 VRx and VRy to A0 and A1
- joystick 2 VRx and VRy to A2 and A3

- joystick buttons to D2 and D3 (not used right now)
- nano D1 (TX) to the red wire of the cable, nano GND to the green wire

the diode has a stripe on one side, that side faces the switch. if you put it in backwards nothing turns on, which is better than frying the nano but still annoying.



my remote, front side



back side with the battery holder

step 3: solder the arm board

this one has more stuff on it cuz every servo gets its own lm7805. servos pull a lot of current when they start moving, and if they all share one supply the uno just resets randomly.

- battery + to diode, then the switch. this is the main line, it goes to uno VIN and to the IN pin of all 4 regulators
- on each 7805 put a 0.01uF cap from IN to ground and a 0.1uF from OUT to ground, right next to the pins
- each 7805 OUT only goes to its own servos red wire, dont join the outputs together
- all servo brown wires, the 7805 grounds, uno GND and battery - all go to one common ground
- servo signal (orange) wires: base to D3, shoulder to D5, elbow to D6, gripper to D9
- cable red wire to uno D0 (RX), cable green wire to uno GND
- another led and 330 ohm on the uno 5V as a power light

i used female berg strips for the servo plugs and the uno so nothing needs desoldering later. the full schematics are in the PCB folder.



the arm board ends up under the base plate

step 4: test both boards

seriously do this. i had a cold joint on mine and finding it now took 5 minutes, finding it later would have taken forever.

1. with no battery in, check there is no short between the main line and ground (continuity mode on the multimeter)
2. put the battery in and flip the switch, the leds should light up
3. check each 7805 output is around 5v
4. check the nano 5V pin is around 5v and the joystick pins change voltage when you move the sticks
5. upload the arm code with SERVO_TEST set to 1, plug in the servos and check they sweep one by one

important: D0 on the uno is shared with the usb. unplug the red wire from D0 every time you upload and plug it back after, otherwise the upload fails. took me like 4 failed uploads to figure that out.

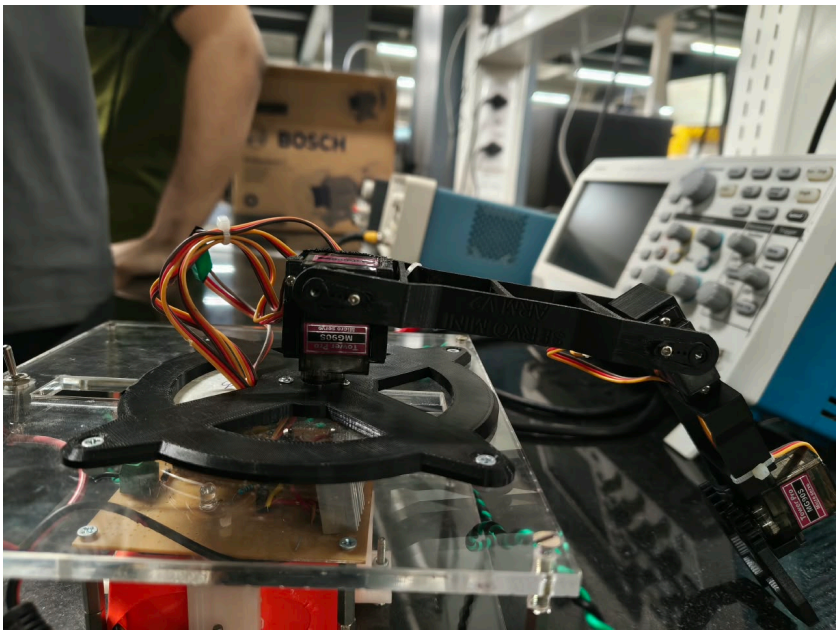
step 5: center the servos before putting the horns on

power the servos with the normal arm code (SERVO_TEST set to 0). when it starts it moves every servo to its start angle, 90 for base, shoulder and elbow and 45 for the gripper. now put the horns on so each joint is in the middle of its range at that position.

if you skip this a joint can hit its end way too early and youll have to take half the arm apart again. ask me how i know.

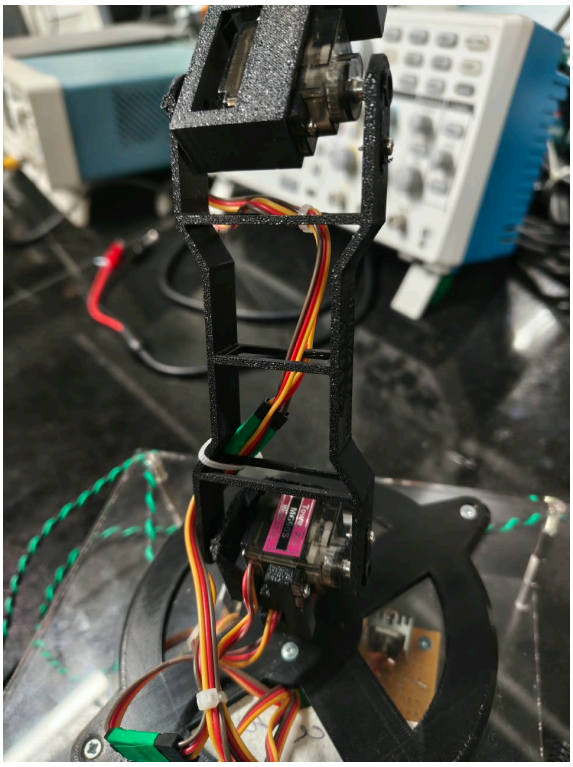
step 6: build the base

1. screw the m3 spacers into the acrylic base
2. put the arm board (with the uno plugged in) on the spacers
3. put the battery and switch somewhere you can still reach
4. screw the printed base plate on top of the spacers
5. mount the base servo in the middle of the plate, shaft facing up



step 7: bracket, arm 1 and arm 2

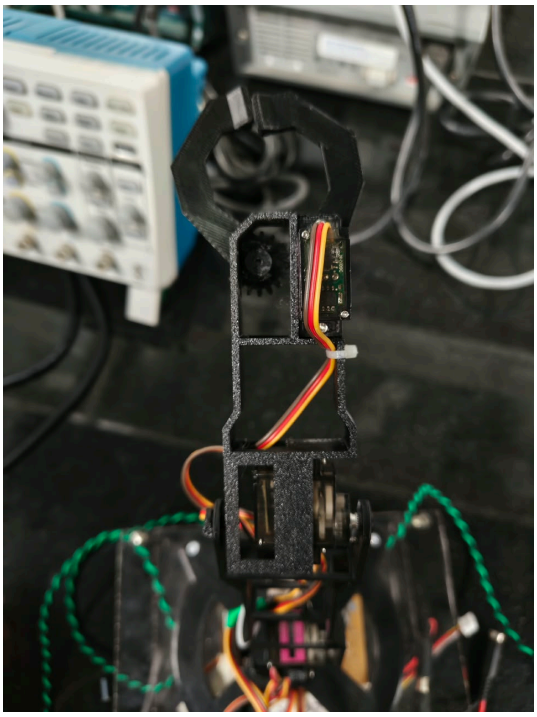
1. push the shoulder servo into the bracket and bolt it. the bottom of the bracket goes on the base servo horn so the whole thing rotates
2. attach arm 1 to the shoulder servo horn on one side and the pivot on the other
3. slide the elbow servo into the pocket at the top of arm 1, then bolt arm 2 on
4. with the power off, move each joint by hand. nothing should rub



arm 1 with the shoulder servo at the bottom and the elbow servo at the top

step 8: gripper

1. put the gripper servo in its pocket at the end of arm 2
2. put the driving gear on the servo horn and the second gear on its pin so they mesh
3. put on both fingers and close it up with the top cover
4. at 45 degrees the claw should be about half open. if not, take the gear off and put it back one tooth over



step 9: wires

4 servos worth of wires on a spinning base gets messy really fast. run them through the holes in the arm and zip tie them along the links. leave some slack near the base and at every joint, then move every joint all the way and check no wire gets pulled or pinched.

step 10: code

1. upload Tx_Nano_Joystick to the nano
2. open the serial monitor at 9600 baud. you should see <0,0,0,0> with the sticks in the middle, and numbers up to 255 or -255 when you push them
3. upload Rx_Uno_Servo to the uno (SERVO_TEST 0, RATE_MODE 1), with the red wire off D0
4. connect the cable, turn on the arm first and then the remote
5. move the sticks slowly and check every joint moves

step 11: tuning

all the settings are at the top of the arm code:

- MIN_ANG and MAX_ANG: move each joint slowly to its ends and set these so nothing hits anything. my gripper is 10 to 80
- INVERT: set to true for any joint that moves the opposite way to the stick
- MAX_SPEED_DEG_S: how fast the joints move, 90 felt right to me
- DEBUG: set to 1 to print what the uno is getting, really helps when something is off

in rate mode the stick controls the speed and not the position, so when you let go the joint just stays there. thats why the claw keeps holding the cube.

if something goes wrong

- serial monitor shows 255 on everything: the nano is reading max on every pin, check the joystick header wiring
- upload fails on the uno: red wire is still on D0
- arm doesnt move at all: check the green wire is connected and the red wire is on D0 not D1. turn on DEBUG and see if it says LINK OK
- arm drifts on its own: make DEADBAND in the remote code a bit bigger
- servo jitters or the uno resets: check the caps are close to the 7805 pins, add a 100uF cap on that servos 5v, check the battery
- joint goes the wrong way: flip it in INVERT
- joint stops before the stick is fully pushed: horn wasnt centered, go back to step 5
- 7805 gets hot: kinda normal going from 11.1v down to 5v, a small heatsink helps if its too hot to touch

done!

thats it, you should have a working robot arm now :D try something small and light first, i used a 20 mm cube around 5-10 g. the whole build story with all my mistakes is in JOURNAL.md in the repo. if something here doesnt make sense open an issue and ill fix it.